FLATNESS OF THE AUGMENTATION MAP IS EQUIVALENT TO FLATNESS OF ALL ITS SIMPLICIAL LEVELS

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-flat-augmentation-modules (source lines 2285-2294); the proof environment begins two lines below it and is its own proof, closing at source line 2323.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the total site, the level morphisms and the structure sheaves are developed earlier in the same chapter.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-spaces-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Spaces chapter, counted once: the augmentation map is flat exactly when all of its simplicial levels are flat.

Allowed standard prerequisites (editorial).

- Flat morphism of ringed topoi. A morphism of ringed topoi is flat when its structure sheaf is flat as a sheaf of modules over the inverse image of the structure sheaf of the target.. The conclusion of the converse direction is exactly this condition, checked on modules (source lines 2290-2293, 2300-2303).
- Simplicial site and its total site. The total site of a simplicial site has as objects the objects of all levels, and the level morphisms g_n : C_n -> C_total are flat, with a_n = a composed with g_n.. It supplies the flatness of g_n used in the easy direction and the identification a_n = a composed with g_n used in the converse (source lines 2297-2300, 2320-2322).
- Inverse image of sheaves of modules along a composite. The inverse image along a composite of morphisms of topoi is the composite of the inverse images, so g_n^{-1} a^{-1} equals a_n^{-1} on structure sheaves.. It is the identification that turns the flatness hypothesis on a_n into the required injectivity (source lines 2320-2322).

Lemma 0.1. *With notation as above. The morphism* $a : (Sh(\mathcal{C}_{total}), \mathcal{O}) \to (Sh(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$ *is flat if and only if* $a_n : (Sh(\mathcal{C}_n), \mathcal{O}_n) \to (Sh(\mathcal{D}), \mathcal{O}_{\mathcal{D}})$ *is flat for* $n \geq 0$.

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Proof. Since $g_n : (Sh(\mathcal{C}_n), \mathcal{O}_n) \to (Sh(\mathcal{C}_{total}), \mathcal{O})$ is flat, we see that if a is flat, then $a_n = a \circ g_n$ is flat as a composition. Conversely, suppose that a_n is flat for all n. We have to check that $\mathcal{O}$ is flat as a sheaf of $a^{-1}\mathcal{O}_\mathcal{D}$-modules. Let $\mathcal{F} \to \mathcal{G}$ be an injective map of $a^{-1}\mathcal{O}_\mathcal{D}$-modules. We have to show that

$$\mathcal{F} \otimes_{a^{-1}\mathcal{O}_\mathcal{D}} \mathcal{O} \to \mathcal{G} \otimes_{a^{-1}\mathcal{O}_\mathcal{D}} \mathcal{O}$$

is injective. We can check this on $\mathcal{C}_n$, i.e., after applying g_n^{-1}. Since $g_n^* = g_n^{-1}$ because $g_n^{-1}\mathcal{O} = \mathcal{O}_n$ we obtain

$$g_n^{-1}\mathcal{F} \otimes_{g_n^{-1}a^{-1}\mathcal{O}_\mathcal{D}} \mathcal{O}_n \to g_n^{-1}\mathcal{G} \otimes_{g_n^{-1}a^{-1}\mathcal{O}_\mathcal{D}} \mathcal{O}_n$$

which is injective because $g_n^{-1}a^{-1}\mathcal{O}_\mathcal{D} = a_n^{-1}\mathcal{O}_\mathcal{D}$ and we assume a_n was flat. $\square$